

Computer System Design

Lab 1: Working with Remote Linux Systems using SSH

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Indian Institute of Technology Bhilai

Learning Outcomes

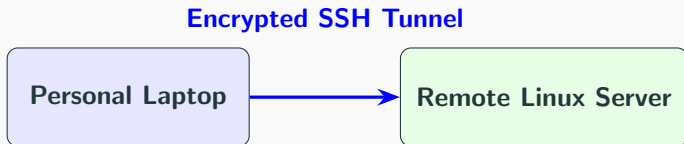
- Understand SSH and remote systems
- Connect to allocated Linux machines
- Install required packages
- Execute commonly used Linux commands
- Submit a command execution report

SSH

Why SSH?

SSH (Secure Shell) enables encrypted communication with remote Linux systems.

- Cloud servers
- HPC clusters
- AI/ML GPUs
- Remote development
- DevOps



Connecting to a Remote System

General Syntax

```
ssh [OPTIONS] username@host_name_or_IP_Address
```

Common Usage

```
# Default SSH (Port 22)
ssh default@127.0.0.1

# Connect to a custom port
ssh -p 2222 default@127.0.0.1

# Authenticate using a private key
ssh -i ~/.ssh/id_ed25519 \
    default@127.0.0.1

# Both together
ssh -i ~/.ssh/id_ed25519 \
    -p 2222 default@127.0.0.1
```

Option	Purpose
-p	Custom SSH port
-i	Private key file
-v	Verbose output
-X	X11 forwarding
-L	Local port forwarding

First Login

Verify the server fingerprint, type yes, then authenticate.

Allocated Machines

Finding Your Systems

1. Open the emailed Allocation Sheet.
2. Find your Name / Roll number.
3. Note your four assigned systems as well as their SSH command.
4. SSH into your system and change password when prompted in first login.
5. Re-login with your changed password

NOTE : All logins can be monitored so refrain for accessing others' machines or messing with their passwords

System Details

1. No of systems allotted per Student : 4
2. RAM per system : 512MB
3. vCPU cores per system : 1
4. Storage per system : 1.5GB

NOTE : Most Tools and Packages are already present inside each system along with Ubuntu and would not require installing.

Environment Setup

Installed Packages

The following tools are already installed in each system:

```
# Development
```

```
git build-essential python3 python3-pip conda
```

```
# Networking
```

```
openssh-server curl wget net-tools iproute2 iputils-ping
```

```
# Utilities
```

```
tree htop unzip ca-certificates
```

```
# Python Libraries
```

```
numpy pandas scipy matplotlib scikit-learn
```

```
torch torchvision torchaudio jupyter ipython tqdm
```

1. Use the conda base env if installing any python package, do not create a new venv
2. In need of any linux packages/commands, use apt package installer
3. Refrain from using heavy packages or installations

Useful Commands

System Information & Hardware

```
whoami          # Current logged-in user
id              # User ID, Group ID
hostname        # Machine hostname
hostnamectl     # OS and hostname details
uname -a        # Kernel and system information

lscpu           # CPU information
free -h         # Memory usage
df -h           # Disk usage
lsblk           # Storage devices
```

Networking & Process Management

```
ifconfig                # Show IP addresses
ping google.com         # Test connectivity
ss -tuln                # Open TCP/UDP ports
curl ifconfig.me        # Public IP address
wget URL                # Download a file

ps aux                  # Running processes
top                     # Live process monitor
                        # (Ctrl+C to exit this command)
htop                    # Interactive process monitor
                        # (Ctrl+C to exit this command)
kill <PID>              # Terminate a process
                        # (Carefull which process to kill)
```

Files, Archives & Git

```
find .                # Search files
grep "text" file.txt  # Search inside files
history               # Command history
tree                  # Directory tree

tar -czvf out.tar.gz dir/  # Create archive
tar -xzvf out.tar.gz      # Extract archive

git clone URL          # Clone repository
git status              # Repository status
git pull                # Fetch latest changes
```

Useful Tools

```
pwd                # Current directory
which python3      # Locate executable
which conda        # Locate if conda present
which ssh          # Locate SSH client

history            # Previous commands
alias              # User-defined shortcuts

nohup cmd &        # Run in background
jobs               # Background jobs
```

Exercise

Lab Exercise & Report Submission

Lab Exercise

1. Find your assigned system from the allocation sheet.
2. Connect using SSH.
3. Install the required packages.
4. Execute all demonstrated commands.
5. Save screenshots of every output.
6. Logout using `exit`.

Report Submission

- Student Name
- Roll Number
- Assigned Systems (IP/Hostname + Port)
- Commands Executed
- Relevant Screenshots

Submission

Submit a single PDF report in Google Classroom/Form before the deadline.

Common Errors & Best Practices

Common Errors

- **Permission denied:** Incorrect credentials or file permissions.
- **Connection refused:** SSH service not running or wrong port.
- **Host unreachable:** Network or IP address issue.
- **sudo Authentication failure:** Incorrect password or privileges.
- **Package not found:** Run `sudo apt update` first.

Best Practices

- Never share your SSH password or private keys.
- Logout after completing your work using `exit`.
- Avoid modifying or terminating other users' systems.
- Look into SSH keys for frequent or automated access.
- **Loss of System access will cause data loss when reinstating system**

Thank You!

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